



Pavlo Hrab

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Summary

PhD candidate in Marine Microbiome Bioinformatics (WUR) with expertise at the interface of microbiology and computational science. Experienced in developing large-scale multi-omics pipelines to support cultivation strategies and applying machine learning to study bacterial lifestyles. Skilled in combining wet-lab isolation with computational analyses and in translating results across interdisciplinary teams

Work experience

Postdoctoral researcher,

Wageningen University and Research
Bioinformatics group; Marnix Medema group

Jan 2026 - Sep 2026

Wageningen, Netherlands

PhD Candidate,

Wageningen University and Research
Bioinformatics group & Microbiology group
Supervisors: Detmer Sipkema; Marnix Medema

Jan 2022 - Dec 2025,

Wageningen, Netherlands

Project: Meta'omic analysis of sponge microbiota

- Planned and led a **1-year long** uninterrupted bacteria isolation study in unconventional conditions, leading towards uncovering new physiological properties in isolates
- Built and analyzed **metagenomic and metatranscriptomic datasets** to characterize sponge-associated microbial communities and guide cultivation strategies
- Communicated my work in **7 multi-disciplinary** conferences, across microbiology, bioinformatics, and natural products fields

Technical Support Specialist,

Information System Development

May 2021 - Oct 2021,

Lviv, Ukraine

Education

MSc, Ivan Franko National University of Lviv

Major - Genetics and Biotechnology.
(diploma cum laude)

Sep 2019 - Dec 2020

Lviv, Ukraine

BSc, Ivan Franko National University of Lviv

Major - Biology.

Sep 2015 - Jun 2019

Lviv, Ukraine

Professional skills

Laboratory skills	Cultivation and isolation of bacteria; PCR; DNA cloning.
Computational skills	R; Python; bash; workflow automation (nextflow); phylogenetics; pan- and metagenomics; (meta)transcriptomics; functional annotation; machine learning; big data analysis.
Interpersonal skills	Course lecturing and practical teaching; mentoring BSc and MSc students; public speaking at international conferences; scientific writing; project coordination

Publications

1. Jackson, S. A., **Hrab, P.**, Zdouc, M. M., Clarke, D. J., & Dobson, A. D. W. (2026). New insights into the microbiome of the deep-sea sponge *Inflatella pellicula* and the secondary metabolic potential of metagenome-assembled genomes and the wider microbiome. *Microbial genomics*, 12(1), 001602. <https://doi.org/10.1099/mgen.0.001602>
2. Zdouc, M. M., Blin, K., ... **Hrab P.**, ... Medema, M. H. (2025). MIBiG 4.0: advancing biosynthetic gene cluster curation through global collaboration. *Nucleic acids research*, 53(D1), D678–D690. <https://doi.org/10.1093/nar/gkae1115>
3. **Hrab, P.**, Rückert, C., Busche, T., Ostash, I., Kalinowski, J., Fedorenko, V., Yushchuk, O., & Ostash, B. (2021). Complete genome sequence of *Streptomyces cyanogenus* S136, producer of anticancer angucycline landomycin A. *3 Biotech*, 11(6), 282. <https://doi.org/10.1007/s13205-021-02834-4>

Highlighted conference talks

1. **Pavlo Hrab**, Mitja Zdouc, Hans-Joachim Ruscheweyh, Shinichi Sunagawa, Detmer Sipkema and Marnix Medema “Marine Acidobacteriota as halogenation reservoirs: Discovery of novel biosynthetic gene clusters”, ECMNP 2025, Piran, **Slovenia**, September 29 - October 3, 2025 (oral talk)
2. **Pavlo Hrab**, Loureiro C., Schorn M., Sipkema D., Medema M. “Unraveling the *Geodia barretti* microbiome biosynthetic potential through the lens of cultivation”, Natural Products (Meta) Genome Mining, Copenhagen, **Denmark**, May 30-31, 2022 (oral talk)

Awards and Honors

Summer School for Natural Products research
August 28, 2025, Ukraine

Invited speaker

Languages

English (advanced, C1), **Ukrainian** (native).

Hobbies

Reading, water polo, and multiday trekking.